

HYBRID FRAUD DETECTION SYSTEM

TEAM : APEX FORGE

PROBLEM STATEMENT

USERS LACK A RELIABLE, REAL-TIME WAY TO IDENTIFY AND UNDERSTAND SCAM MESSAGES, LEADING TO FINANCIAL AND DATA LOSS—OUR SOLUTION BRIDGES THIS GAP USING EXPLAINABLE AI.

OUR OBJECTIVE

TO BUILD AN AI-POWERED SCAM DETECTION SYSTEM THAT:

**ANALYZES MESSAGES IN REAL-TIME
IDENTIFIES SUSPICIOUS PATTERNS (KEYWORDS, LINKS, INTENT)**

**CLASSIFIES MESSAGES AS SAFE OR SCAM
PROVIDES ACTIONABLE INSIGHTS AND WARNINGS TO PROTECT USERS**

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- 3. PROVIDES ACTIONABLE INSIGHTS AND WARNINGS TO
PROTECT USERS**

SOLUTION APPROACH

STEP 1: INPUT MESSAGE , USER COPY PASTES THE SMS OR TEXT AND CLICKS ON ANALYZE



AI Scam Message Detector

Paste any message and check if it's SAFE or SCAM.

Enter message:

Dear Customer, aapka Airtel Axis Bank Credit Card upgrade ke liye qualify ho gaya hai. 20,000 reward points redem karne ke liye link par click karein: [fake-link]. 24 hrs mein expire ho jayega

Analyze 

STEP 2: LANGUAGE DETECTION
LANGDETECT → IDENTIFIES LANGUAGE

STEP 3: TRANSLATION
NON-ENGLISH → ENGLISH USING GOOGLE TRANSLATOR

STEP 4: FEATURE EXTRACTION
REGEX FINDS:
UPI IDS
LINKS
AMOUNTS

Detected Entities

```
{  
  "upi_ids" : []  
  "links" : []  
  "amounts" : []  
}
```

STEP 5: ML PREDICTION
COUNTVECTORIZER CONVERTS TEXT
NAIVE BAYES PREDICTS:
SCAM / SAFE
CONFIDENCE SCORE

STEP 6: RULE-BASED RISK ENGINE
KEYWORD SCORING SYSTEM
LINK DETECTION
FINAL RISK SCORE

STEP 7: DECISION ENGINE
HIGH / MEDIUM / LOW RISK CLASSIFICATION

 **Result**

 **HIGH RISK (75)**

AI Prediction: SCAM

STEP 8: EXPLAINABILITY LAYER
REASONS LIST
HIGHLIGHTED WORDS
ADVICE GENERATION

STEP 9: LOGGING SYSTEM
SAVES RESULTS INTO CSV

STEP 10: OUTPUT DISPLAY
CLI OUTPUT (PRETTY_PRINT)
OR STREAMLIT DASHBOARD (UI VERSION)



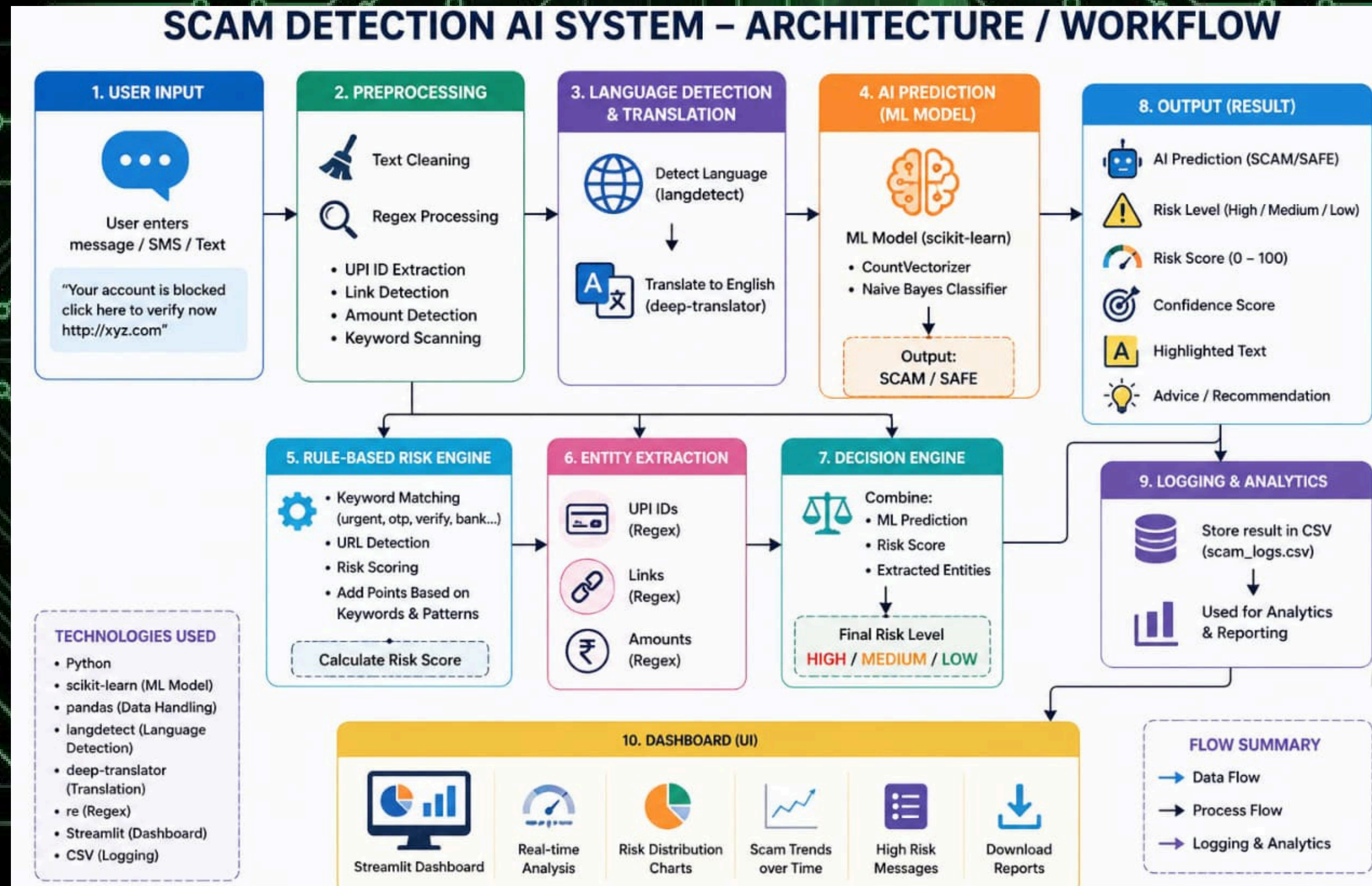
Advice: Do NOT click links or send money.

Reason: Keyword detected: kyc, Keyword detected: account, Keyword detected: blocked, Keyword detected: bank, Keyword detected: update

Highlighted Text:

Dear Customer, apka [BANK] [ACCOUNT] block hone wala hai! [KYC] document pending hai. Please immediately [UPDATE] karein: bit.ly/bank-kyc

ARCHITECTURE / WORKFLOW



TECH STACK

**FRONTEND
STREAMLIT**

**BACKEND
PYTHON**

**AI/ML
SCIKIT-LEARN (NAIVE BAYES)**

**NLP
LANGDETECT, DEEP-TRANSLATOR**

**DATA PROCESSING
PANDAS, REGEX**

DEMO SCREENSHOTS



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Analyze 🚀

🔍 Result 🔗

🚨 HIGH RISK (35)

AI Prediction: SCAM

📊 Details

Details

```
{
  "risk": {
    "level": "HIGH"
    "score": 75
    "confidence": 0.75
  }
  "ai_prediction": {
    "label": "SCAM"
    "confidence": 1
  }
}
```

Insights

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 Latest Logs Preview

	timestamp	text
0	2026-04-24T22:25:16.782865	urgent do ur kyc
1	2026-04-24T22:28:43.445786	Dear Customer, apka Bank Account block hone wala hai! KYC document pending

FUTURE SCOPE

Browser extension

Mobile app (SMS scanner)

Upgrade to deep learning (BERT)

API for banks / fintech apps